



```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv(r'C:\Users\HP\dataset_1_202606151013.csv')
```

```
In [5]: df
```

```
Out[5]:
```

	destination	passanger	weather	temperature	time	coupon
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House
...	...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away
12681	Work	Alone	Snowy	30	7AM	Coffee House
12682	Work	Alone	Snowy	30	7AM	Bar
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)

12684 rows × 7 columns

```
In [8]: df[['weather', 'temperature']]
```

Out[8]:

	weather	temperature
0	Sunny	55
1	Sunny	80
2	Sunny	80
3	Sunny	80
4	Sunny	80
...	...	...
12679	Rainy	55
12680	Rainy	55
12681	Snowy	30
12682	Snowy	30
12683	Sunny	80

12684 rows × 2 columns

In [9]: `df.head(10)`

Out[9]:

	destination	passanger	weather	temperature	time	coupon	expirat
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0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	

10 rows × 27 columns

```
In [ ]: df['passanger'].unique()
```

```
In [ ]: df[sql['destination']=='Home']
```

```
In [ ]: df.sort_values('coupon')
```

```
In [ ]: df.rename(columns={'destination':'Destination'},inplace=True)
```

```
In [ ]: df
```

```
In [ ]: df.groupby('occupation').size().to_frame('Count').reset_index()
```

```
In [ ]: df.groupby('weather')['temperature'].mean().to_frame('avg_temp').reset_index()
```

```
In [ ]: df.groupby('weather')['temperature'].size().to_frame('Count_temp').reset_index()
```

```
In [ ]: df.groupby('weather')['temperature'].nunique().to_frame('count_distinct_temp')
```

```
In [12]: df.groupby('weather')['temperature'].sum().to_frame('sum_temp').reset_index()
```

```
Out[12]:
```

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

```
In [13]: df.groupby('weather')['temperature'].min().to_frame('min_temp').reset_index()
```

```
Out[13]:
```

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	30

```
In [14]: df.groupby('weather')['temperature'].max().to_frame('max_temp').reset_index()
```

```
Out[14]:
```

	weather	max_temp
0	Rainy	55
1	Snowy	30
2	Sunny	80

```
In [15]: df.groupby('occupation').filter(lambda x: x['occupation'].iloc[0] == 'Student')
```

```
Out[15]: occupation
Student    1584
dtype: int64
```

```
In [28]: pd.concat([df, df])['destination'].drop_duplicates()
```

```
Out[28]: 0    No Urgent Place
13           Home
16           Work
Name: destination, dtype: object
```

```
In [38]: pd.merge(df, df[['time', 'part_of_day']], on='time', how='inner')[['destination', 'time', 'part_of_day']]
```

```
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NameError                                Traceback (most recent call last)
Cell In[38], line 1
----> 1 pd.merge(df, df1[['time', 'part_of_day']], on='time', how='inner')
[['destination', 'time', 'part_of_day']]

NameError: name 'df1' is not defined
```

```
In [23]: df[df['passanger'] == 'Alone'][['destination', 'passanger']]
```

Out[23]:

	destination	passanger
0	No Urgent Place	Alone
13	Home	Alone
14	Home	Alone
15	Home	Alone
16	Work	Alone
...	...	...
12676	Home	Alone
12680	Work	Alone
12681	Work	Alone
12682	Work	Alone
12683	Work	Alone

7305 rows × 2 columns

In [24]: `df[df['weather'].str.startswith('Sun')]`

Out[24]:

	destination	passanger	weather	temperature	time	coupon
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House
...	...	...	...	...	...	...
12673	Home	Alone	Sunny	30	6PM	Carry out & Take away
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)

10069 rows × 27 columns

```
In [27]: df[(df['temperature'] >= 29) & (df['temperature'] <= 75)][['temperature']].unique
```

```
Out[27]: array([55, 30])
```

```
In [26]: df[df['occupation'].isin(['Sales & Related', 'Management'])][['occupation']]
```

```
Out[26]:
```

	occupation
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193	Sales & Related
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194	Sales & Related
------------	-----------------

195	Sales & Related
------------	-----------------

196	Sales & Related
------------	-----------------

197	Sales & Related
------------	-----------------

...	...
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12679	Sales & Related
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12680	Sales & Related
--------------	-----------------

12681	Sales & Related
--------------	-----------------

12682	Sales & Related
--------------	-----------------

12683	Sales & Related
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1931 rows × 1 columns